The Schrödinger Equation as Operator Evolution

A bridge from vectors and matrices to wavefunctions

Why quantum mechanics looks like linear algebra in an infinite-dimensional space

How to read this tutorial

Pause at each boxed result and ask what it says geometrically or physically. The derivation is part of the explanation, not a hurdle to skip.

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1 From coordinate lists to functions

A finite vector may be written

$$\boldsymbol{v} = (v_1, v_2, \ldots, v_N).$$

The index i tells us which coordinate we are reading. A function $\psi(x)$ can be viewed similarly: the continuous label x replaces the discrete label i.

The finite inner product

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \sum_i u_i^* v_i$$

becomes

$$\boxed{\langle \phi, \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x)\psi(x)\, \mathrm{d}x.}$$

The norm is

$$\|\psi\|^2 = \langle \psi, \psi \rangle = \int |\psi(x)|^2\, \mathrm{d}x.$$

The bridge

A wavefunction is not merely a graph in physical space. It is one vector in a function space. Its value at x is analogous to a coordinate of that vector in the position-labelled description.

2 Normalisation and probability

Quantum states are normalised so that

$$\boxed{\int_{-\infty}^{\infty} |\psi(x,t)|^2\, \mathrm{d}x = 1.}$$

For an interval $[a, b]$,

$$\Pr(a \le x \le b) = \int_a^b |\psi(x,t)|^2\, \mathrm{d}x.$$

Multiplying ψ by a global phase $\mathrm{e}^{i\alpha}$ does not change $|\psi|^2$, so state vectors that differ only by a global phase represent the same physical ray.

3 Operators are the infinite-dimensional analogue of matrices

A matrix maps vectors to vectors. A linear operator maps functions to functions:

$$\widehat{A}(\alpha\psi + \beta\phi) = \alpha\widehat{A}\psi + \beta\widehat{A}\phi.$$

Important examples in one dimension are

$$\widehat{x}\psi = x\psi, \qquad \widehat{p}\psi = -i\hbar\frac{\partial\psi}{\partial x}.$$

The Hamiltonian for a particle in a potential is

$$\boxed{\widehat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x).}$$

The kinetic energy is a differential operator; the potential energy is a multiplication operator.

4 Self-adjointness and real measurements

An operator is self-adjoint when

$$\langle \phi, \widehat{A}\psi \rangle = \langle \widehat{A}\phi, \psi \rangle$$

for all functions in an appropriate domain.

If

$$\widehat{A}\psi = a\psi,$$

then, for a normalised eigenfunction,

$$\begin{aligned} a &= \langle \psi, a\psi \rangle = \langle \psi, \widehat{A}\psi \rangle \\ &= \langle \widehat{A}\psi, \psi \rangle = a^*. \end{aligned}$$

Hence a is real.

Domains are part of the operator

A differential expression alone is not a complete operator. Boundary conditions and regularity determine the domain, and the domain determines whether integration by parts produces unwanted boundary terms.

5 The time-dependent Schrödinger equation

The evolution law is

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = \widehat{H}\psi.}$$

In vector notation,

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t} |\psi(t)\rangle = \widehat{H} |\psi(t)\rangle .$$

This is a first-order differential equation in time for a vector in a function space. Compare it with a finite linear system

$$\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = A\boldsymbol{v}.$$

The Hamiltonian is the generator of quantum time evolution.

If $\widehat{H}$ is time independent, the formal solution is

$$\boxed{|\psi(t)\rangle = exp\left(-\frac{i}{\hbar}\widehat{H}t\right)|\psi(0)\rangle .}$$

6 Why the norm is preserved

Let $\widehat{H}$ be self-adjoint. Then

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\langle \psi, \psi \rangle &= \left\langle \frac{\partial \psi}{\partial t}, \psi \right\rangle + \left\langle \psi, \frac{\partial \psi}{\partial t} \right\rangle \\ &= \frac{i}{\hbar}\langle \widehat{H}\psi, \psi \rangle - \frac{i}{\hbar}\langle \psi, \widehat{H}\psi \rangle = 0. \end{aligned}$$

Thus a normalised state stays normalised. The evolution operator is unitary.

7 Energy eigenfunctions separate the time dependence

Suppose

$$\widehat{H}\phi_n = E_n\phi_n.$$

Then

$$\psi_n(x,t) = \phi_n(x)\mathrm{e}^{-iE_nt/\hbar}$$

solves the time-dependent equation. Its probability density is stationary:

$$|\psi_n(x,t)|^2 = |\phi_n(x)|^2.$$

For a superposition,

$$\psi(x,0) = \sum_n c_n\phi_n(x),$$

linearity gives

$$\boxed{\psi(x,t) = \sum_n c_n\mathrm{e}^{-iE_nt/\hbar}\phi_n(x).}$$

Time evolution rotates each energy coordinate by its own phase.

8 The infinite square well

Let

$$V(x) = 0 \quad (0 < x < L), \qquad V = \infty \quad \text{outside.}$$

The boundary conditions are

$$\phi(0) = \phi(L) = 0.$$

Inside the well,

$$-\frac{\hbar^2}{2m}\phi'' = E\phi.$$

Write

$$k^2 = \frac{2mE}{\hbar^2}.$$

Then

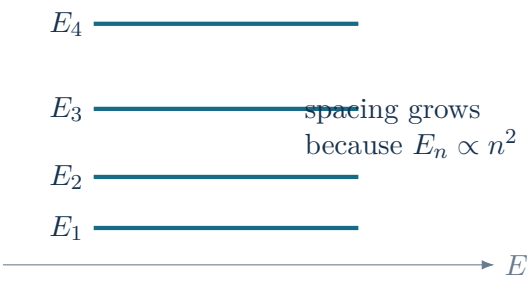
$$\phi(x) = A\sin(kx) + B\cos(kx).$$

The condition at $x = 0$ gives $B = 0$. The condition at $x = L$ gives

$$\sin(kL) = 0 \quad \Longrightarrow \quad kL = n\pi.$$

Therefore

$$\boxed{\phi_n(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}, \qquad E_n = \frac{\hbar^2\pi^2n^2}{2mL^2}.}$$



Quantisation is a boundary-condition effect: only certain standing-wave shapes fit the interval.

9 Fourier coordinates diagonalise free evolution

For a free particle,

$$\widehat{H} = -\frac{\hbar^2}{2m}\partial_x^2.$$

The plane wave e^{ikx} is an eigenfunction because

$$\partial_x^2 \mathrm{e}^{ikx} = -k^2 \mathrm{e}^{ikx}.$$

Thus

$$\widehat{H}\mathrm{e}^{ikx} = \frac{\hbar^2 k^2}{2m}\mathrm{e}^{ikx}.$$

In Fourier coordinates, the differential operator becomes multiplication by $\hbar^2k^2/(2m)$. Each Fourier component evolves independently:

$$\widetilde{\psi}(k,t) = \mathrm{e}^{-i\hbar k^2 t/(2m)}\widetilde{\psi}(k,0).$$

10 Heat flow and quantum evolution

The heat equation

$$\frac{\partial u}{\partial t} = D\partial_x^2 u$$

and the free Schrödinger equation

$$\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m}\partial_x^2 \psi$$

use the same spatial operator but different time generators.

For a Fourier mode e^{ikx}:

$$\widetilde{u}(k,t) = \mathrm{e}^{-Dk^2 t}\widetilde{u}(k,0),$$

so high-frequency heat modes decay. Quantum modes acquire phases:

$$\widetilde{\psi}(k,t) = \mathrm{e}^{-i\hbar k^2 t/(2m)}\widetilde{\psi}(k,0),$$

so their magnitudes do not decay.

Same basis, different geometry

Heat flow contracts vectors toward smoother states. Quantum evolution rotates state-space coordinates without changing the norm.

11 Commutators and incompatible operations

The commutator is

$$[\widehat{A},\widehat{B}] = \widehat{A}\widehat{B} - \widehat{B}\widehat{A}.$$

For position and momentum,

$$\begin{aligned}[\widehat{x},\widehat{p}]\psi &= x(-i\hbar\psi') + i\hbar(x\psi)' \\ &= i\hbar\psi.\end{aligned}$$

Hence

$$\boxed{[\widehat{x},\widehat{p}] = i\hbar\widehat{I}.}$$

Noncommutation is the operator statement behind the position–momentum uncertainty relation.

12 Expectation values

For a normalised state,

$$\boxed{\langle A\rangle = \langle \psi, \widehat{A}\psi\rangle = \int \psi^*(x)(\widehat{A}\psi)(x)\,\mathrm{d}x.}$$

The spread is

$$(\Delta A)^2 = \langle (\widehat{A} - \langle A\rangle)^2\rangle.$$

The expectation is a weighted average over possible measurement outcomes, not generally an eigenvalue attached to the current state.

13 What to remember

1. A wavefunction is a vector in a function space.
2. Integrals replace finite coordinate sums in inner products.
3. Differential operators play the role of matrices.
4. Self-adjointness supplies real eigenvalues and unitary evolution.
5. The Hamiltonian generates time evolution.
6. Choosing an eigenbasis can turn an operator equation into independent scalar equations for coordinates.

The infinite-dimensional bridge

The conceptual jump from linear algebra to quantum mechanics is smaller than it first appears: replace finite coordinate lists by functions, sums by integrals, matrices by operators, and matrix exponentials by operator exponentials.